Objective: Learn basic SQL commands using MySQL.

Task: Create a database named college_db and a table named students.

Create a database named college_db.

```
mysql> CREATE DATABASE college_db;
Query OK, 1 row affected (0.04 sec)

mysql> USE college_db;
Database changed
```

Create a table named students with the following columns:

```
mysql> CREATE TABLE students (
    -> student_id INT PRIMARY KEY
    -> AUTO_INCREMENT,
    -> name VARCHAR(100),
    -> age INT,
    -> course VARCHAR(50),
    -> city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

Insert at least 5 student records into the table with different names, ages, courses, and

cities.

```
mysql> INSERT INTO students(name, age, course, city)
    -> VALUES
    -> ('Murali', 21, 'BCA', 'Bengaluru'),
    -> ('vinod', 22, 'BCA', 'Hosur'),
    -> ('Prasad', 20, 'BCom', 'Bengaluru'),
    -> ('Hari', 23, 'BCA', 'chennai'),
    -> ('Jeevan', 19, 'BSC', 'Mysuru');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

1. Display all student records.

```
mysql> SELECT * FROM students;
+------------+--------+------+--------+-----------+
| student_id | name   | age  | course | city      |
+------------+--------+------+--------+-----------+
|          1 | Murali |   21 | BCA    | Bengaluru |
|          2 | vinod  |   22 | BCA    | Hosur     |
|          3 | Prasad |   20 | BCom   | Bengaluru |
|          4 | Hari   |   23 | BCA    | chennai   |
|          5 | Jeevan |   19 | BSC    | Mysuru    |
+------------+--------+------+--------+-----------+
5 rows in set (0.00 sec)
```

2. Display only the student names and courses.

```
mysql> SELECT name, course FROM students;
+--------+--------+
| name   | course |
+--------+--------+
| Murali | BCA    |
| vinod  | BCA    |
| Prasad | BCom   |
| Hari   | BCA    |
| Jeevan | BSC    |
+--------+--------+
5 rows in set (0.00 sec)
```

3. Find students whose age is greater than 20.

```
mysql> SELECT * FROM students
    -> WHERE age > 20;
+------------+--------+------+--------+-----------+
| student_id | name   | age  | course | city      |
+------------+--------+------+--------+-----------+
|          1 | Murali |   21 | BCA    | Bengaluru |
|          2 | vinod  |   22 | BCA    | Hosur     |
|          4 | Hari   |   23 | BCA    | chennai   |
+------------+--------+------+--------+-----------+
3 rows in set (0.02 sec)
```

4. Find students who belong to Bengaluru.

```
mysql> SELECT * FROM students
    -> WHERE city = 'bengaluru';
+------------+--------+------+--------+-----------+
| student_id | name   | age  | course | city      |
+------------+--------+------+--------+-----------+
|          1 | Murali |   21 | BCA    | Bengaluru |
|          3 | Prasad |   20 | BCom   | Bengaluru |
+------------+--------+------+--------+-----------+
2 rows in set (0.00 sec)
```

5. Display students studying BCA.

```
mysql> SELECT * FROM students
    -> WHERE course = 'BCA';
+------------+--------+------+--------+-----------+
| student_id | name   | age  | course | city      |
+------------+--------+------+--------+-----------+
|          1 | Murali |   21 | BCA    | Bengaluru |
|          2 | vinod  |   22 | BCA    | Hosur     |
|          4 | Hari   |   23 | BCA    | chennai   |
+------------+--------+------+--------+-----------+
3 rows in set (0.00 sec)
```

6. Display all students in ascending order of age.

```
mysql> SELECT * FROM students
    -> ORDER BY age ASC;
+------------+--------+------+--------+-----------+
| student_id | name   | age  | course | city      |
+------------+--------+------+--------+-----------+
|          5 | Jeevan |   19 | BSC    | Mysuru    |
|          3 | Prasad |   20 | BCom   | Bengaluru |
|          1 | Murali |   21 | BCA    | Bengaluru |
|          2 | vinod  |   22 | BCA    | Hosur     |
|          4 | Hari   |   23 | BCA    | chennai   |
+------------+--------+------+--------+-----------+
5 rows in set (0.00 sec)
```

7. Update one student's city.

```
mysql> UPDATE students
    -> SET city = 'Chennai'
    -> WHERE name = 'Murali';
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM students;
+------------+--------+------+--------+-----------+
| student_id | name   | age  | course | city      |
+------------+--------+------+--------+-----------+
|          1 | Murali |   21 | BCA    | Chennai   |
|          2 | vinod  |   22 | BCA    | Hosur     |
|          3 | Prasad |   20 | BCom   | Bengaluru |
|          4 | Hari   |   23 | BCA    | chennai   |
|          5 | Jeevan |   19 | BSC    | Mysuru    |
+------------+--------+------+--------+-----------+
5 rows in set (0.00 sec)
```

8. Count the total number of students.

```
mysql> SELECT COUNT(*)AS total_students
    -> FROM students;
+---------------+
| total_students |
+---------------+
|             5 |
+---------------+
1 row in set (0.03 sec)
```

9. Delete one student record using their student ID.

```
mysql> DELETE FROM students
    -> WHERE student_id = 3;
Query OK, 1 row affected (0.04 sec)

mysql> SELECT * FROM students;
+------------+--------+------+--------+---------+
| student_id | name   | age  | course | city    |
+------------+--------+------+--------+---------+
|          1 | Murali |   21 | BCA    | Chennai |
|          2 | vinod  |   22 | BCA    | Hosur   |
|          4 | Hari   |   23 | BCA    | chennai |
|          5 | Jeevan |   19 | BSC    | Mysuru  |
+------------+--------+------+--------+---------+
4 rows in set (0.00 sec)
```